

Sensitivity analysis

1 Introduction

Suppose we have found an optimal solution to a LP, but now something in the data changes, or we realize that the data was inaccurate. Examples of such occurrences are:

- change in the price of one of the products we buy or sell;
- change in the availability of a raw material used in the production;
- a new product can be manufactured;
- a new constraint appears.

We also could want to analyze the robustness of our solution to changes in the market. In all these cases, we want to:

- Decide if the current solution is still optimal;
- If it is not, find an optimal solution faster than solving the new LP from scratch.

We will next investigate how to achieve the goals above in the various cases. Given a feasible solution to an LP, we assume that we have a fast algorithm that can check if it is optimal and, if it is, compute an optimal solution for the dual. These assumptions are justified by the fact that solvers can perform these operations very fast.

As a running example, we will use (a variation of) the production problem often seen during the course. A company produces two kinds of goods, chairs and tables:

- It takes 1 unit of wood and 2 units of paint to build a table.
- It takes 1 unit of wood and 1 unit of paint to build a chair.
- Each week, we have 80 units of wood and 100 units of paint.
- The profit for producing and selling a table is \$3.
- The profit for producing and selling a chair is \$2.

Our goal is to decide the weekly production as to maximize the profit. This problem can be formulated as the LP

$$\begin{aligned} \max \quad & 3x_1 + 2x_2 \\ & x_1 + x_2 \leq 80 \\ & 2x_1 + x_2 \leq 100 \\ & x_1, x_2 \geq 0, \end{aligned} \quad (\text{P})$$

where x_1 is the number of tables produced weekly, x_2 is the number of chairs produced weekly, the first constraint makes sure that we do not use more wood than we have, and the second constraint makes sure that we do not use more paint than we have. Its optimal solution is $x^* = (20, 60)$. It will also be useful to work with its dual

$$\begin{aligned} \min \quad & 80y_1 + 100y_2 \\ & y_1 + 2y_2 \geq 3 \\ & y_1 + y_2 \geq 2 \\ & y_1, y_2 \geq 0 \end{aligned} \quad (\text{D})$$

whose optimal solution is $y^* = (1, 1)$. Remember that y_1^* can be interpreted as the price of one unit of wood on the market; and y_2^* can be interpreted as the price of one unit of paint on the market.

2 Adding a variable or a constraint

2.1 Manufacturing a new product (adding a new variable)

Suppose that we can now produce a third product – say, windows. The amount of windows we produce will be denoted by x_3 . For instance, the new LP could become:

$$\begin{aligned} \max \quad & 3x_1 + 2x_2 + 3x_3 \\ & x_1 + x_2 + 2x_3 \leq 80 \\ & 2x_1 + x_2 + 3x_3 \leq 100 \\ & x_1, x_2, x_3 \geq 0 \end{aligned}$$

To find an optimal solution to the new problem, we proceed as follows.

- The optimal solution x^* can be “completed” by adding a 0 in x_3 (that is, no window is produced). The production vector we obtain—call it x' —is clearly feasible for the new problem, and we can use the solver to check if it is optimal.
- If not, use x' as a starting point to warm-start the solution to new LP. The solver can then start from x' as a feasible point and will not have to solve the LP from scratch. This is especially valuable in very large linear programs.

2.2 Adding a new (production) constraint

Now suppose we have a new constraint on the number of nails needed to produce chairs and tables. For instance, the new LP could look as follows:

$$\begin{aligned} \max \quad & 3x_1 + 2x_2 \\ & x_1 + x_2 \leq 80 \\ & 2x_1 + x_2 \leq 100 \\ & 20x_1 + 40x_2 \leq 800 \\ & x_1, x_2 \geq 0 \end{aligned} \quad (\text{P}')$$

To solve the new LP, we can:

- Check if x^* is still feasible. If it is, it must be optimal.
- Else, x^* is infeasible and we cannot use it as a starting point for the solution of the new LP. However, adding a constraint to an LP corresponds to adding a variable to its dual, which now looks as follows:

$$\begin{array}{rcl}
 \min & 80y_1 + 100y_2 + 800y_3 & \\
 & y_1 + 2y_2 + 20y_3 \geq 3 & \\
 & y_1 + y_2 + 40y_3 \geq 2 & (D') \\
 & y_1, y_2, y_3 \geq 0 &
 \end{array}$$

- Similar to what was done (for the primal) in Section 2.1, we can find an optimal solution $\bar{y}$ for (D') using as a starting solution the optimal solution $y' = (1, 1, 0)$ for the original dual – notice that y' is y^* , to which a 0 has been added in the last component.
- Since the dual of (D') is (P'), from $\bar{y}$ we can easily find an optimal solution to (P').

3 Change in the objective function or in the right-hand side

3.1 Change in the availability of a good (right-hand side)

We want to address next the following question: Will the optimal production plan change if the amounts of available wood changes? If it does, how much more profit can we make?

Increase in the availability of wood. Let us start with small variations – say, from 80 to 81 units. In this case:

- x^* is still feasible, and gives the same profit as in the original problem. But may not be optimal – that is, there may be a newly feasible solution x' which gives a larger profit.
- y^* is still feasible for the dual, since the constraints of the dual have not changed. The objective function value of y^* changes by 1, since the coefficient of y_1 in the objective function increases by 1, and $y_1^* = 1$. Since we assumed small variations, we can assume that y^* is still optimal for the dual.
- By strong duality, there is an optimal solution to the primal that leads to 1 more unit of profit – and that's optimal for the new primal. We can either decide to keep the original production plan x^* , or compute a new optimal production plan to obtain an increase of profit of 1.

Note that for larger increases of the amount of available wood—say, 10, 50, 100—it is less likely that y^* will stay optimal. Thus, while y^* stays feasible for the new dual, and increases its objective function value by 10, 50, 100 respectively, there may be another feasible solution to the dual, say y' that is optimal in the new dual. Thus, y' has smaller cost in the new dual than y^* . We deduce that the increase of profit we can get is *at most* 10, 50, 100 in each of the cases above.

Decrease in the availability of wood. Let us now consider the case when the availability of wood decreases from 80 to 79.

- x^* is not feasible anymore, so we will have to compute a new production plan.
- Assuming that y^* stays optimal (since the variation in the right-hand side is small), we can repeat the argument above and conclude that the new optimal production plan gives 1 less profit.

If the amount of wood decreases by more—say, 5, 10, 100—then y^* will still be feasible but may not be optimal. The new optimal production will decrease its profit by *at least* 5, 10, 100.

Increase in the availability of wood redux. Consider the following modification of the primal problem (the amount of available wood is now 150) and its dual

$$\begin{array}{ll}
 \max & 3x_1 + 2x_2 \\
 & x_1 + x_2 \leq 150 \\
 & 2x_1 + x_2 \leq 100 \\
 & x_1, x_2 \geq 0 \\
 \min & 150y_1 + 100y_2 \\
 & y_1 + 2y_2 \geq 3 \\
 & y_1 + y_2 \geq 2 \\
 & y_1, y_2 \geq 0
 \end{array}$$

Its optimal solutions are $\bar{x} = (0, 100)$ and $\bar{y} = (0, 2)$, respectively. What happens if, starting from this new problem, we increase or decrease the units of available wood by 1? In both cases,

- $\bar{x}$ stays feasible for the primal.
- Since the variation is small, we can assume that $\bar{y}$ stays optimal for the dual. However, the objective function cost of $\bar{y}$ will not change (since $\bar{y}_1 = 0$), so $\bar{x}$ is still optimal for the primal. In other words, the change on the availability of wood has no effect. This makes sense since $\bar{x}$ uses fewer units of wood than we had available (or in other words, the first constraint is not tight at $\bar{x}$), thus small changes in the availability of wood will not affect us.

It is important to remark that the discussion above only works since we are assuming small variations of the right-hand side.

Changes in the right-hand side: the general case. We can give a general version of the discussion from this section. Let x^*, y^* be optimal solutions for a primal and dual in the form:

$$\begin{array}{ll}
 \max & c^T x \\
 (\hat{P}) & Ax \leq b \\
 & x \geq 0 \\
 \min & b^T y \\
 (\hat{D}) & A^T y \geq c \\
 & y \geq 0
 \end{array}$$

If the right-hand side of the i -th constraint of $(\hat{P})$ changes¹ by Δ small enough, then the objective function value of the optimum of $(\hat{P})$ will change by $y_i^* \cdot \Delta$.

A similar discussion can be repeated for LPs in a form different from the one above. However, you can also always transform a generic LP in an equivalent LP that has the form of $(\hat{P})$ above.

3.2 Changing the profits (objective function)

We now address the question of whether the optimal solution changes if we change one coefficient in the objective function. We are going to assume that this coefficient is that of x_1 , but a similar argument can be made for the change of any coefficient (as long as only one coefficient changes).

The production problem. Let us consider again the original production problem seen throughout the notes, which we report, together with its dual:

¹ We are making here the implicit assumption that the primal and dual feasible region is non-degenerate. Many solvers perturb their instances as to make sure they are non-degenerate, so it is a reasonable assumption. We will not deal with degenerate polytopes in this class.

$$\begin{array}{ll}
\max & 3x_1 + 2x_2 \\
& x_1 + x_2 \leq 80 \\
& 2x_1 + x_2 \leq 100 \\
& x_1, x_2 \geq 0,
\end{array} \quad (P)
\qquad
\begin{array}{ll}
\min & 80y_1 + 100y_2 \\
& y_1 + 2y_2 \geq 3 \\
& y_1 + y_2 \geq 2 \\
& y_1, y_2 \geq 0.
\end{array} \quad (D)$$

Assume that c_1 (the objective function coefficient of x_1) changes from 3 to some value $\hat{c}_1$. What should we expect to happen?

- Since the constraints have not changed, x^* is still feasible for the primal. Its profit has now changed by $(\hat{c}_1 - 3) \cdot x_1^* = 20\hat{c}_1 - 60$, since $c_1 = 3$ has changed to $\hat{c}_1$. x^* may however not be optimal anymore.
- We want to understand for which value of $\hat{c}_1$ x^* stays optimal. Intuitively, we expect that for values of $\hat{c}_1$ close to 3 it stays indeed optimal, while for values that are further away it will not.

Let us now compute the values of $\hat{c}_1$ for which x^* stays optimal. To do that, recall how we constructed the dual: we found multipliers for the constraints of (P) so that the resulting inequality was an upper bound to the objective function of (P). The multiplier we obtained are exactly the optimal solution to (D). Indeed, if we multiply

$$x_1 + x_2 \leq 80 \quad (1)$$

by $y_1^* = 1$ and

$$2x_1 + x_2 \leq 100 \quad (2)$$

by $y_2^* = 1$, we obtain

$$3x_1 + 2x_2 \leq 180.$$

Note that x^* lies at the intersection of the equations given by (1), (2). Thus, x^* will stay an optimal solution to the modified (P) as long as there exists *nonnegative* multipliers y_1 and y_2 that, multiplied by the left hand side of (1) and (2), respectively, lead to $\hat{c}_1 x_1 + 2x_2$. We can express the latter requirement compactly as

$$y_1 \cdot (1, 1) + y_2 \cdot (2, 1) = (\hat{c}_1, 2)$$

or as a system of equations as

$$\begin{array}{rcl}
y_1 + 2y_2 & = & \hat{c}_1 \\
y_1 + y_2 & = & 2.
\end{array} \quad (3)$$

The system above has two variables y_1 and y_2 , one parameter $\hat{c}_1$, and two equations. It has the unique solution

$$\begin{array}{rcl}
\hat{y}_1 & = & 4 - \hat{c}_1 \\
\hat{y}_2 & = & \hat{c}_1 - 2.
\end{array} \quad (4)$$

Recall that we want the multipliers to be nonnegative. Thus, imposing nonnegativity on $\hat{y}_1$ and $\hat{y}_2$, we obtain the conditions

$$\begin{array}{rcl}
4 - \hat{c}_1 & \geq & 0 \\
\hat{c}_1 - 2 & \geq & 0
\end{array}$$

or equivalently

$$\boxed{2 \leq \hat{c}_1 \leq 4}$$

This is the interval for which x^* stays optimal for the primal. As a sanity check, we can see that $c_1 = 3$ belongs to this interval – as it must! So no change in our production plan is needed if the profit for selling table fluctuates by 1\$ at most.

A dual perspective on the production problem. Let us consider the argument above from the point of view of the dual. What we want is to construct a solution $\hat{y} = (\hat{y}_1, \hat{y}_2)$ of the dual that certifies the optimality of x^* . For that, we need $\hat{y}$ to be feasible and to satisfy complementary slackness with x^* . Let us discuss each of these conditions:

- a) Complementary slackness between primal variables and dual constraints: Since x_1^*, x_2^* are both strictly greater than 0, $\hat{y}$ satisfy the two constraints of the dual at equality. Equivalently, $\hat{y}$ must be a solution to the system (3).
- b) Complementary slackness between dual variables and primal constraints: Since (1) and (2) are satisfied at equality by x^* , both $\hat{y}_1$ and $\hat{y}_2$ are allowed to be different from 0. So no further restriction on $\hat{y}$ is imposed.
- c) Feasibility. $A^T \hat{y} \geq c$ is implied by a). Nonnegativity of $\hat{y}_1, \hat{y}_2$ leads to (4).

Thus, the two approaches lead to the same solution.

The production problem with an additional constraint. We modify (P) by adding one more constraint as follows:

$$\begin{aligned} \max \quad & \hat{c}_1 x_1 + 2x_2 \\ & x_1 + x_2 \leq 80 \\ & 2x_1 + x_2 \leq 100 \\ & x_1 + 3x_2 \leq 240 \quad (P'') \\ & x_1, x_2 \geq 0. \end{aligned}$$

We can observe that $x^* = (20, 60)$ is still optimal for the new problem. However, x^* does not satisfy it at equality: $20 + 3 \cdot 60 = 200 < 240$. Thus, we want only to use the multiplier corresponding to the first two inequalities (which are satisfied at equality by x^*) to generate the objective function. This leads again to the system (3), which has the same solutions. Hence, once again x^* stays optimal for

$$\boxed{2 \leq \hat{c}_1 \leq 4}$$

Now let us consider the dual perspective on this problem. The dual of P'' is:

$$\begin{aligned} \min \quad & 80y_1 + 100y_2 + 240y_3 \\ & y_1 + 2y_2 + y_3 \geq \hat{c}_1 \\ & y_1 + y_2 + 3y_3 \geq 2 \quad (D'') \\ & y_1, y_2, y_3 \geq 0. \end{aligned}$$

in the variables y_1, y_2, y_3 . We want to certify the optimality of x^* by constructing a solution $\hat{y} = (\hat{y}_1, \hat{y}_2, \hat{y}_3)$ of the dual that certifies the optimality of x^* . For that, we need $\hat{y}$ to be feasible and to satisfy complementary slackness with x^* . Let us discuss each of these conditions. We underline the parts that are different from the example without the additional constraint.

- a) Complementary slackness between primal variables and dual constraints: Since x_1^*, x_2^* are both strictly greater than 0, $\hat{y}$ satisfy the two constraints of the dual at equality. Equivalently, $\hat{y}$ must be a solution to the system (3).
- b) Complementary slackness between dual variables and primal constraints: Since (1) and (2) are satisfied at equality by x^* , both $\hat{y}_1$ and $\hat{y}_2$ are allowed to be different from 0. So no further restriction on $\hat{y}_1, \hat{y}_2$ is imposed. However, the third (new) constraint is not satisfied at equality by x^* . Thus, $\hat{y}_3 = 0$. This is why we do not see y_3 in the system (3).
- c) Feasibility. $A^T \hat{y} \geq c$ is implied by a) and b). Nonnegativity of y_3 is implied by b). Nonnegativity of $\hat{y}_1, \hat{y}_2$ leads to (4).

Thus, the required dual solution is $\hat{y} = (\hat{y}_1, \hat{y}_2, 0)$, where $\hat{y}_1, \hat{y}_2$ are solutions to (3).